

Akshiv Bansal

SOFTWARE ENGINEER

🏠 www.akshiv.com 📱 [akshiv](#) 💻 [akshiv-bansal](#)

Education

University of British Columbia

Vancouver, Canada

BACHELOR OF APPLIED SCIENCE, IN ENGINEERING PHYSICS WITH DISTINCTION

2013-2018

Minor in Honors Mathematics; Dean's Honor List

Experience

Merlin, get-merlin.com

San Francisco, CA

CO-FOUNDER

Oct 2025 - Mar 2026

- Working across the stack to design an smart component selection engine for electrical components.
- Used python (FastAPI) as the backend and NextJS as the frontend to create a visual way of comparing capacitors, allowing users to compare key engineering graphs and find drop-in replacements for their parts.
- Worked on SEO, analytics, user login/auth, and billing to build out the product.

Humane, Inc. (Acquired by HP)

San Francisco, CA

LEAD SOFTWARE ENGINEER

Jul 2022 - Oct 2025

- Designed and deployed a build artifact management solution used across six different firmware projects, to automate the hardware in the loop testing of built firmware artifacts
- Wrote several Java+Android based applications for Factory Automation, to accomplish tasks like: automatic camera calibration, subsystem co-existence testing, IMU sensor bring-up. Eventually, those tools were also adopted by cross-functional partners including: CVML, EE, RF, and System Software teams.
- Led the rebuild of the hardware testing lab. Worked with IT and maintenance to provision additional electrical, physical, and networking infrastructure. Migrated hardware test machines to Infrastructure as Code (Ansible), and scaled up test fixtures from 5 to 20.
- Owned the bring up and deployment of several hardware in the loop testbeds for custom hardware, which plugged into Humane's CI/CD system. Through this process triaged and fixed numerous breaking bugs in the operating system code as well as the application stack.
- Working inside of AOSP to allow for custom sensors and hardware, while improving multi-process debug-ability and introspection.
- Lead a cloud service migration to switch vendors, accomplished with zero downtime, no external changes, and resulted in a 80% cost savings.
- Improved CI times for multiple firmware and python projects, reducing runtime by 60%.
- Mentored junior test and infrastructure engineers. Designed and implemented good coding standards for the wider team. Helped hire for key roles in the Quality and Build/Release Orgs.

Symbio Robotics

Emeryville, CA

SENIOR SOFTWARE ENGINEER, FRAMEWORK

Nov 2021 - May 2022

- Worked on a Robotic Platform for automated assembly using robots from vendors like ABB and UR
- Worked on logging, data analysis, and applications to understand large volumes of robotic data
- Worked on a force control application involving gear meshing using robots. Implemented the application using async zmq clients and servers

Johnson & Johnson, Robotics and Digital Solutions

Redwood City, CA

SOFTWARE ENGINEER, TOOLS AND INFRASTRUCTURE - ADVANCED DEVELOPMENT

Feb 2020 - June 2021

- Created libraries to provide high level logging capabilities in Python that plugged into low level C++ applications
- Wrote API for the rapid integration of new sensors to expand logging without needing additions to control chain
- Created testing infrastructure to manage test cases and data generated. Worked on streamlined pipeline for data definition, packing, upload, and processing
- Worked with PyTest, Numpy, and Django libraries to build tools for use in automated manufacturing testing